

PRATYUSH NARAIN

Software Engineer 2 | Packet Processing Expertise | Python & CPP Developer

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Summary

Software Engineer with **2.5 years** of experience specializing in **packet processing, DPDK, Linux** based systems and **virtualization**. Holds a degree from **IIT Dhanbad** and has a sound knowledge of **C, C++, Python** data structures, algorithms, git, **low level design** and distributed systems. Has expertise in developing, scalable, performant systems which ensure reliable, secure and available access to customers.

PROFESSIONAL EXPERIENCE

Sandvine

Bangalore, India

Software Engineer

2023 - Present

- **NUMA Aware Load Balancing**
- Implemented NUMA Affine Load Balancing in a 1/2/4 NUMA System for IPV4, IPV6 and GRE/GTPU Tunnel Traffic
- Goal of the feature was to process all the subscriber traffic in a single NUMA node in a multi-socket system and scale it to over 4 Million Subscribers
- Resulted in a 50% improvement on memory consumption due to localisation of shaping objects created per NUMA
- Used concepts of bit masking on IP addresses using 32 and 128 bit spec and masks to create flow rules on 100G mellanox chips in DPDK

Sandvine

Bangalore, India

Software Engineer

2023 - Present

- **DPDK CPU Profiling**
- Feature Lead of implementing a trigger based CPU profiling tool on the engine of the flagship product Active Logic with 256 threads
- Implemented a Linux Perf like algorithm which used context machine addresses and translated them in symbol names finally producing an output of top 10 symbols in a time of 4000 milliseconds
- Resulted in an 80% reduction in turn around time for debugging packet drops

Sandvine

Bangalore, India

Software Engineer

2023 - Present

- **Intent Based Congestion Management**
- Increased network efficiency by 15% by implementing congestion management and intelligent traffic shaping solution
- Developed a solution in Python and C that used raw traffic data like RTT and bandwidth and detected congestion in a particular location
- Handled challenges like Inter-process communication and issues with horizontal scaling when dealing with 1000+ locations

Education

Indian Institute of Technology, Dhanbad

2021 - 2023

M.Tech in Computer Science | CGPA: 8.54 / 10

SKILLS

Languages: C, C++, Python, Bash, JAVA

Engineering: Data Structures, Algorithms, Networking, Load balancing, Linux, System Design, DPDK, SDLC, Software Engineering, Debugging

Tools: Git, GNU (GDB), Wireshark

Projects

Task Offloading in Safety Critical Systems (M.Tech Research)

- Improved system **efficiency** to **83%** by optimizing **dynamic task offloading** to fog nodes in cloud environments
- Implemented **Multi-Armed Bandit** (reinforcement learning) to select the best possible fog node based on past performance using **exploration** and **exploitation**
- Showcased the **value proposition** by comparing the approach against **naive** and **greedy** in an environment of **10 fog nodes and 10,000 tasks**

Key Achievements



Spearheaded a \$3M deployment of IBCM at NOS Portugal

analysing the issues first hand and deciding on key architectural decisions



Have resolved more than 50 critical customer issues



Solved more than 1500 DSA Problems

on platforms like Leetcode and GFG
<https://leetcode.com/u/pratyushnarain5555/>



AIR 546 in GATE 2021 CSE

Achieved a 99.46 percentile



Rated as an excellent performer in my team for the YEAR 2024-25

For outstanding teamwork and innovation